

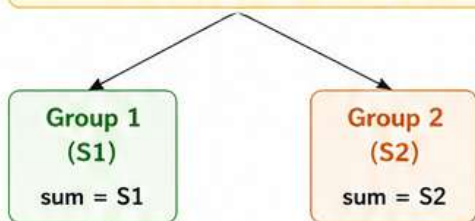
LAST STONE WEIGHT II – LOGIC & DERIVATION

Goal: Minimum possible final stone weight after repeatedly smashing stones.

1. Any sequence of operations can be seen as two groups

Let the stones be: $[a, b, c, d, e, f, \dots]$

After any sequence of smashes,
we can divide all stones into two groups.



Final stone weight = $|S1 - S2|$
(we can assume $S1 \geq S2$)

2. Total sum relation

Let $TOTAL = \text{sum of all stones}$

$$S1 + S2 = TOTAL \quad \dots (1)$$

3. Express S1 in terms of S2

From (1): $S1 = TOTAL - S2 \quad \dots (2)$

4. Write answer in terms of S2

Answer = $S1 - S2$

Using (2):

Answer = $(TOTAL - S2) - S2$

Answer = $TOTAL - 2 * S2 \quad \dots (3)$

5. What do we want to minimize?

$TOTAL$ is fixed.

From (3): Answer = $TOTAL - 2 * S2$

To minimize Answer $\Rightarrow$ Maximize $S2$

6. What is the maximum possible value of S2?

We assumed $S1 \geq S2$

From (1): $S1 = TOTAL - S2$

$S1 \geq S2 \Rightarrow TOTAL - S2 \geq S2$

$\Rightarrow TOTAL \geq 2 * S2$

$\Rightarrow S2 \leq TOTAL / 2 \quad \dots (4)$

7. Combine (3) and (4)

We want to

Maximize $S2$

subject to

$S2 \leq TOTAL / 2$

Therefore,

We need the maximum subset sum
that is as close as possible to $TOTAL / 2$
(without exceeding it).

8. Define the target

Let

TARGET = $\lfloor TOTAL / 2 \rfloor$

(we take floor because subset sum must be $\leq TOTAL/2$)

Our task:

Find maximum subset sum $\leq TARGET$.

9. This is exactly Subset Sum / 0-1 Knapsack

Each stone can be either:

- Take it (put in $S2$)
- Don't take it (put in $S1$)

Each stone can be used at most once $\Rightarrow$ 0-1 Knapsack

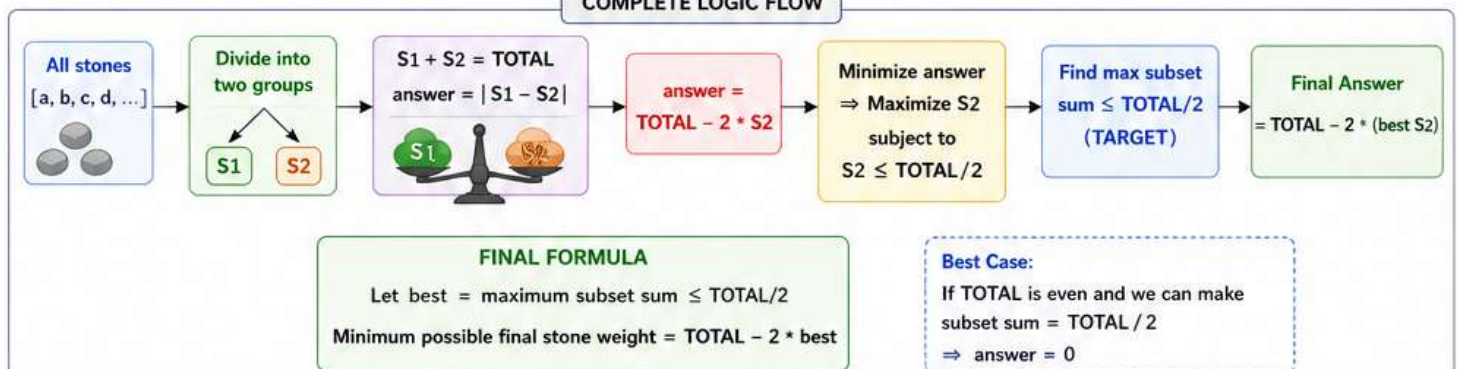
State:

$dp[i][j]$ = maximum subset sum $\leq j$
using first i stones.

Where:

- i = number of stones considered
- j = current capacity ($0 \dots TARGET$)

COMPLETE LOGIC FLOW



[Example:

stones = $[2, 7, 4, 1, 8, 1]$

$TOTAL = 2 + 7 + 4 + 1 + 8 + 1 = 23$

$TARGET = \lfloor 23 / 2 \rfloor = 11$

Maximum subset sum ≤ 11 is 11

(subset: $\{8, 2, 1\}$)



$S2 = 11$

$S1 = 23 - 11 = 12$

Answer = $TOTAL - 2 * S2$

= $23 - 2 * 11$

= 1

LAST STONE WEIGHT II

DP TABULATION - COMPLETE BUILDING

1. LOGIC RECAP

Let stones = [2, 7, 4, 1, 8, 1]

TOTAL = 2 + 7 + 4 + 1 + 8 + 1 = 23

TARGET = TOTAL / 2 = 23 / 2 = 11

Any final answer can be written as:

$$\text{answer} = \text{TOTAL} - 2 * S2$$

To minimize answer → maximize S2

But $S2 \leq \text{TOTAL} / 2 = 11$

Our task :

Find maximum subset sum ≤ 11 .

DP State:

$dp[i][j]$ = maximum subset sum $\leq j$ using first i stones.

(i from 0 ... n, j from 0 ... TARGET)

Transition:

If stones[i-1] > j:

$$dp[i][j] = dp[i-1][j]$$

Else:

$$dp[i][j] = \max(dp[i-1][j], dp[i-1][j - \text{stones}[i-1]] + \text{stones}[i-1])$$

Final Answer:

$$S2 = dp[n][\text{TARGET}]$$

$$\text{Answer} = \text{TOTAL} - 2 * S2$$

Example

Stones = [2, 7, 4, 1, 8, 1]

TOTAL = 23, TARGET = 11

$$S2 = dp[6][11] = 11$$

$$\text{Answer} = 23 - 2 * 11 = 1 \quad \checkmark$$

2. DP TABLE CONSTRUCTION (Row by Row)

Stones (indexed):

Index	0	2	3	4	5
Stone	2	7	4	1	1

Table size = (n+1) x (TARGET+1)

$$= (6+1) \times (11+1) = 7 \times 12$$

Rows → 0 ... 6 (number of stones considered)

Columns → 0 ... 11 (sum limit j)

j →	0	1	2	3	4	5	6	7	8	9	10	11
0 (no stone)	0	0	0	0	0	0	0	0	0	0	0	0
1 (2)	0	0	2	2	2	2	2	2	2	2	2	2
2 (7)	0	0	2	2	2	2	2	7	7	7	9	9
3 (4)	0	0	2	2	4	6	6	7	7	9	10	11
4 (1)	0	1	2	3	4	6	7	8	9	10	11	11
5 (8)	0	1	2	3	4	6	7	8	9	10	11	11
6 (1)	0	1	2	3	4	6	7	8	9	10	11	11

How to read?

$dp[i][j]$ = Max subset sum $\leq j$ using first i stones.

Example:

$dp[3][6] = 6$
(Using first 3 stones [2, 7, 4], maximum sum ≤ 6 is 6 → {2, 4})

Key Observation

Values in each row are non-decreasing when we move right (j increases). Because if a sum is possible for j , it is also possible for any $j+1, j+2, \dots$

3. STEP BY STEP EXPLANATION

Row 0: $i = 0$ (No stone)

We can't take any stone.

So maximum sum for any j is 0.

Row 0 → all 0

j →	0	1	2	3	4	5	6	7	8	9	10	11
$dp[0][j]$	0	0	0	0	0	0	0	0	0	0	0	0

Row 3: $i = 3$ (Stone = 4)

For $j < 4$ → same as above

For $j \geq 4$ → $\max(\text{prev}, \text{prev}[j-4] + 4)$

j →	0	1	2	3	4	5	6	7	8	9	10	11
$dp[3][j]$	0	0	2	2	4	6	6	7	7	9	10	11

Row 6: $i = 6$ (Stone = 1)

For $j < 1$ → same as above

For $j \geq 1$ → $\max(\text{prev}, \text{prev}[j-1] + 1)$

j →	0	1	2	3	4	5	6	7	8	9	10	11
$dp[6][j]$	0	1	2	3	4	6	7	8	9	10	11	11

Row 1: $i = 1$ (Stone = 2)

We can take 2 or not take.

For $j < 2$ → can't take → 0

For $j \geq 2$ → $\max(0, 2) = 2$

j →	0	1	2	3	4	5	6	7	8	9	10	11
$dp[1][j]$	0	0	2	2	2	2	2	2	2	2	2	2

Row 4: $i = 4$ (Stone = 1)

For $j < 1$ → same as above

For $j \geq 1$ → $\max(\text{prev}, \text{prev}[j-1] + 1)$

j →	0	1	2	3	4	5	6	7	8	9	10	11
$dp[4][j]$	0	1	2	3	4	6	7	8	9	10	11	11

Row 2: $i = 2$ (Stone = 7)

We can take 7 or not take.

Check every j :

• For $j < 7$ → same as above

• For $j \geq 7$ → $\max(\text{prev}, \text{prev}[j-7] + 7)$

j →	0	1	2	3	4	5	6	7	8	9	10	11
$dp[2][j]$	0	0	2	2	2	2	2	7	7	7	9	9

Row 5: $i = 5$ (Stone = 8)

For $j < 8$ → same as above

For $j \geq 8$ → $\max(\text{prev}, \text{prev}[j-8] + 8)$

j →	0	1	2	3	4	5	6	7	8	9	10	11
$dp[5][j]$	0	1	2	3	4	6	7	8	9	10	11	11

4. GETTING THE ANSWER

From table:
 $S2 = dp[6][11] = 11$
(maximum subset sum ≤ 11)

$$\text{Remaining sum } S1 = \text{TOTAL} - S2$$

$$= 23 - 11 = 12$$

$$\text{Final answer} = S1 - S2$$

$$= 12 - 11 = 1$$

Using main formula

$$\text{Answer} = \text{TOTAL} - 2 * S2$$

$$= 23 - 2 * 11$$

$$= 1 \quad \checkmark$$

5. INTUITION BEHIND TABLE

- Each row adds one more stone and tries to achieve all possible sums ≤ 11 .
- Each cell chooses the best of two choices:
 - Not take the current stone → $dp[i-1][j]$
 - Take the current stone (if possible) → $dp[i-1][j - \text{stone}] + \text{stone}$
- The value at $dp[n][\text{TARGET}]$ is the best we can do on one side ($S2$).

6. WHY DOES THIS WORK?

- If we can split stones into two groups with sums $S1$ and $S2$, the final result after all operations is $|S1 - S2|$.
- To make $|S1 - S2|$ minimum, the two sums should be as close as possible.
- That happens when $S2$ is maximum and as close as possible to $\text{TOTAL}/2$.